Lecture 05:

Convexity, Convergence, and Optimization

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20 October 2026

Analysing gradient descent

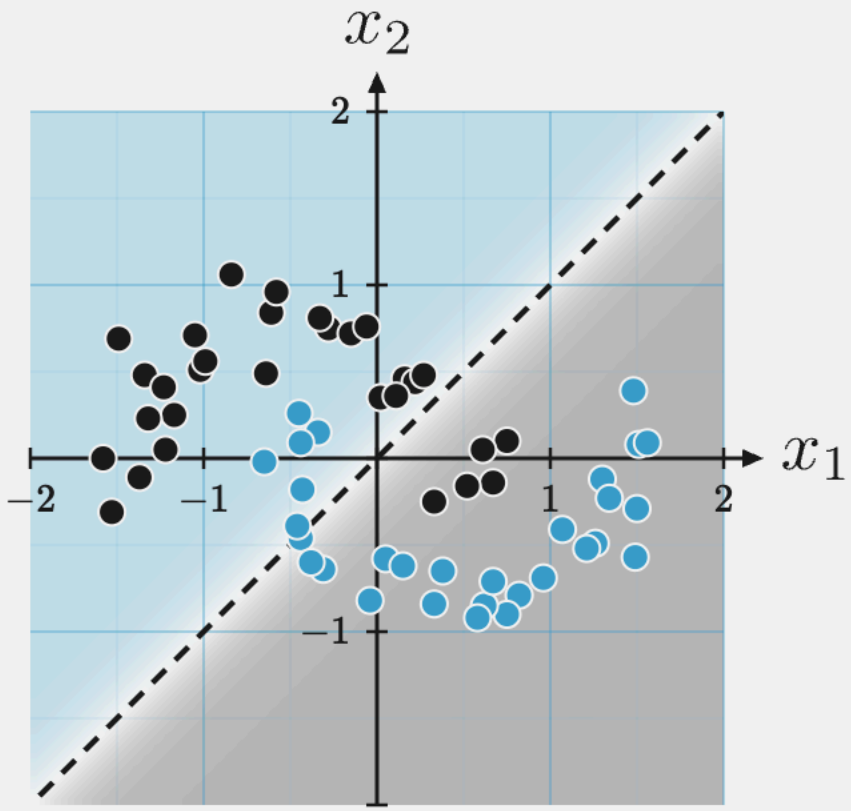
$$\theta^{(t+1)} = \theta^{(t)} - \alpha \nabla_\theta \mathcal{L}(\theta^{(t)})^\top$$

Gradient Descent Algorithm:

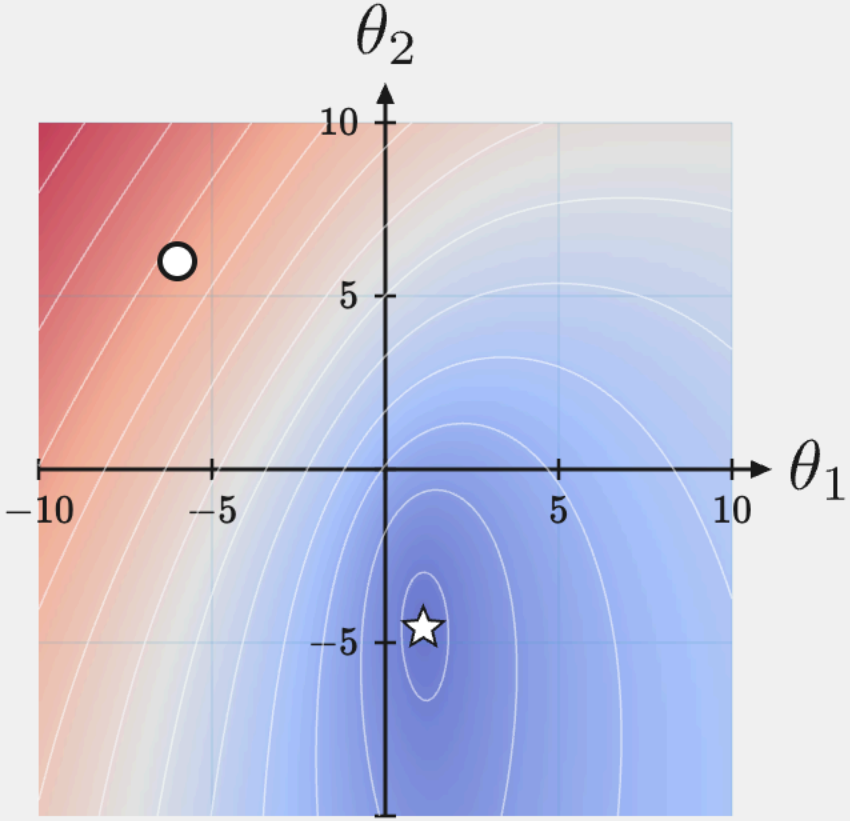
1. Randomly initialize our model
2. Take gradient of loss function w/data
3. Update model parameters w/grad
4. Repeat 2 and 3 '*k*' times

Gradient Descent – When does it work?

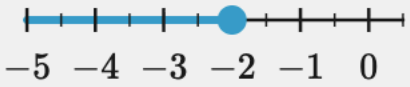
function space: drag or click the data



parameter space



$log_{10}\,\alpha = -2.00$



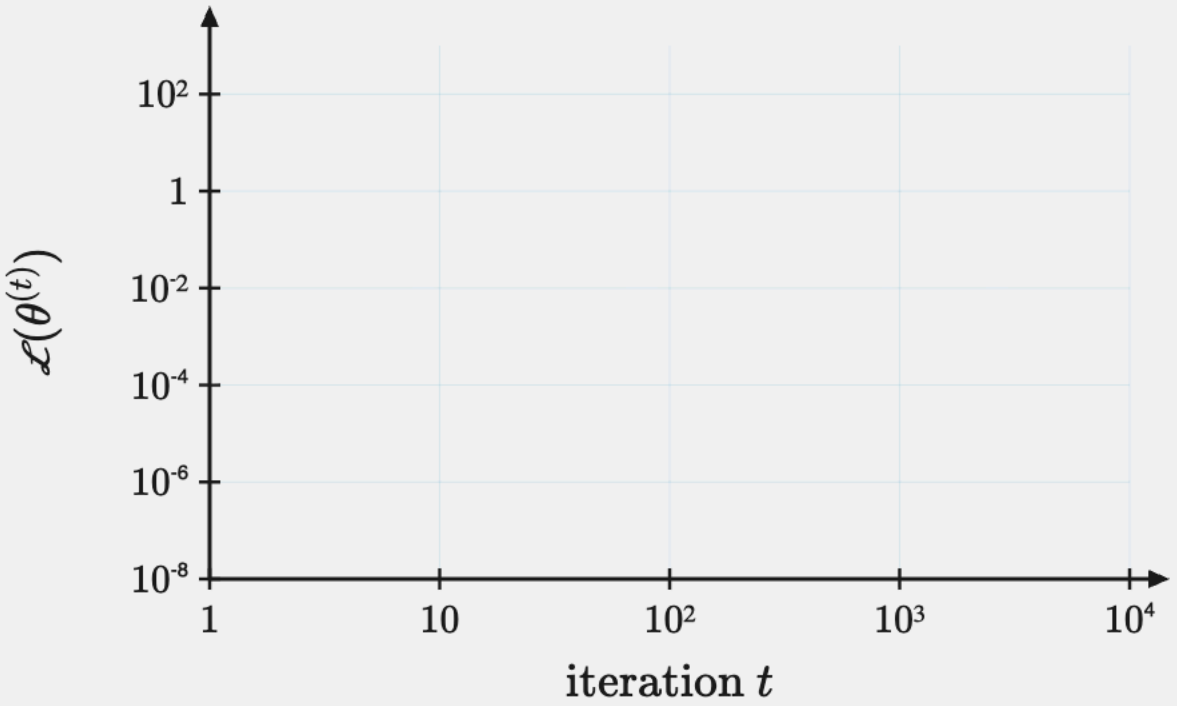
$log_{10}\,k = 3.0$



$t = 0,\ \theta = (-6, 6),\ \mathcal{L} = 362.8,$ accuracy 18%

$\alpha = 0.01,\ k = 1000$

Gradient Descent – When does it work?



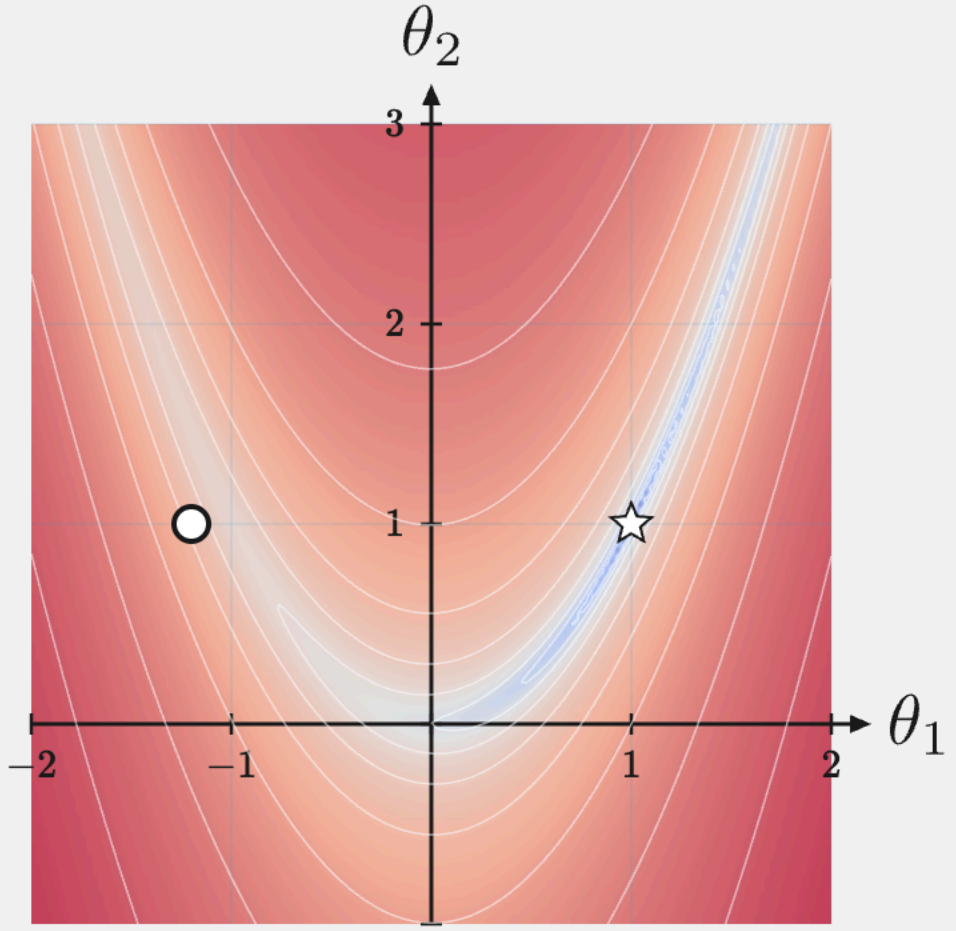
$\alpha = 0.001, \; k = 10000$

$log_{10}\, \alpha = -3.00$



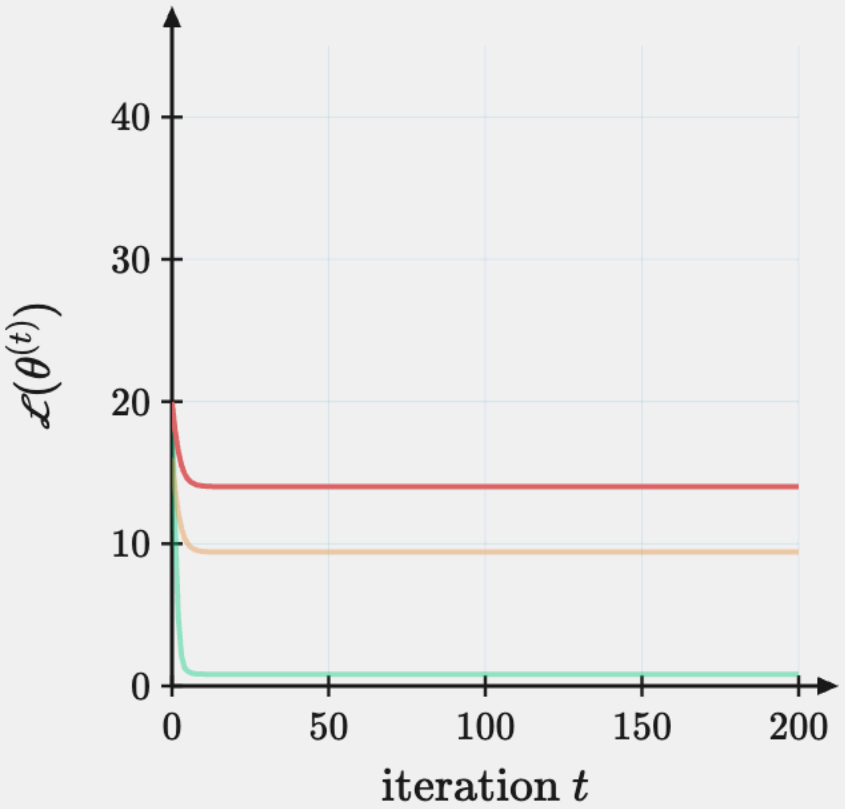
$log_{10}\, k = 4.0$



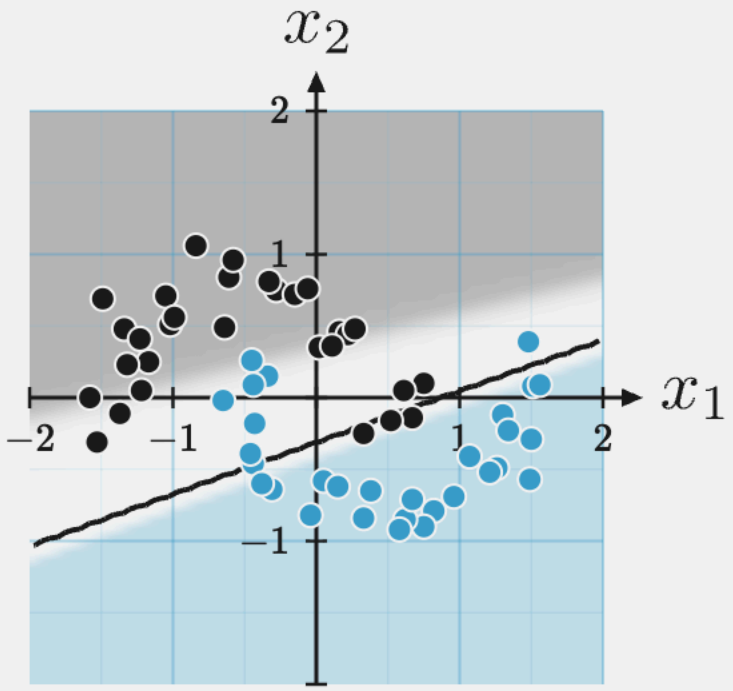


Gradient Descent – When does it work?

loss over the iterations

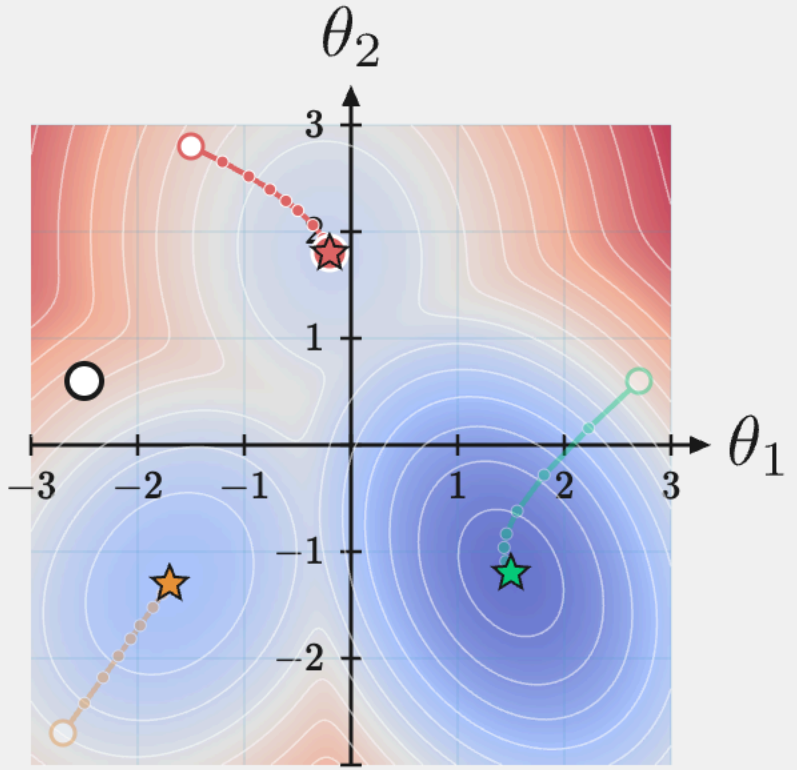


the network at the bottom of this basin



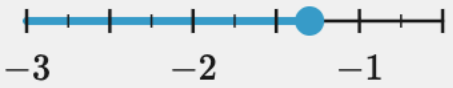
$\mathcal{L} = 14.02$, 82% of the points right

an (artificial) loss landscape



$\alpha = 0.0501$, $k = 200$

$log_{10}\ \alpha = -1.30$



$log_{10}\ k = 2.3$



Analysing gradient descent

$$\theta^{(t+1)} = \theta^{(t)} - \alpha \nabla_\theta \mathcal{L}(\theta^{(t)})^\top$$

Let's begin our analysis of the above algorithm: are we certain that the above equation improves our model?

If you wanted to show that a step of gradient descent "improves" things, what do you want to show?

Analysing gradient descent

$$\theta^{(t+1)} = \theta^{(t)} - \alpha \nabla_\theta \mathcal{L}(\theta^{(t)})^\top$$

Let's begin our analysis of the above algorithm: are we certain that the above equation improves our model?

If you wanted to show that a step of gradient descent "improves" things, what do you want to show?

$$\mathcal{L}(\theta^{(t+1)}) \leq \mathcal{L}(\theta^{(t)}) - c, \qquad c \in \mathbb{R}_+$$

Introducing Assumptions

Proving such a statement with no information about the loss is tough. So we need to introduce assumptions that let us work with the loss.

Assumption 1: a twice-differentiable loss

Proving such a statement with no information about the loss is tough. So we need to introduce assumptions that let us work with the loss.

One tool from calculus we have not used in this course is Taylor expansions which you can see as pointwise approximating polynomial.

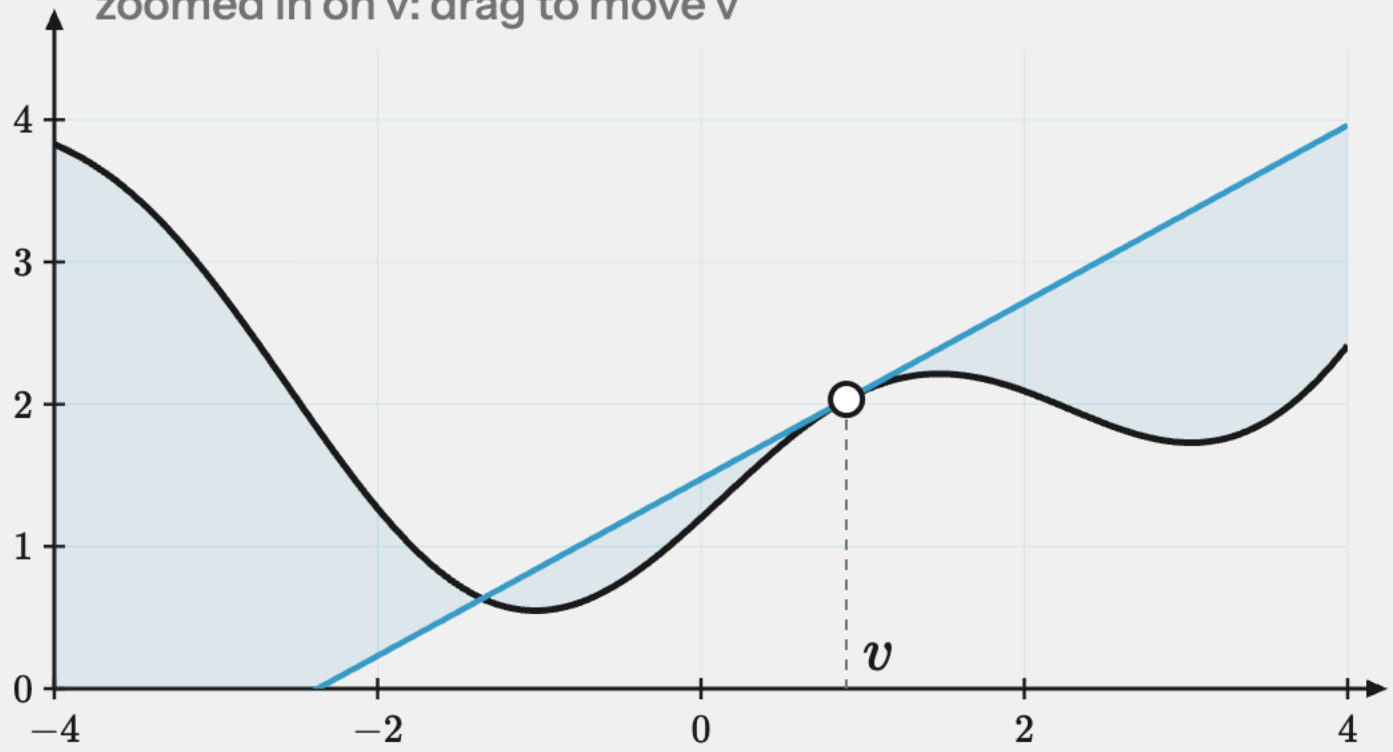
$$\mathcal{L}(\boldsymbol{w}) \approx \mathcal{L}(\boldsymbol{v}) + \underbrace{\nabla \mathcal{L}(\boldsymbol{v})(\boldsymbol{w}-\boldsymbol{v})}_{\text{first order: the gradient}} + \underbrace{\frac{1}{2}(\boldsymbol{w}-\boldsymbol{v})^\top \boldsymbol{H}(\boldsymbol{v})(\boldsymbol{w}-\boldsymbol{v})}_{\text{second order: the Hessian } \boldsymbol{H}(\boldsymbol{v})}$$

Taylor expansion

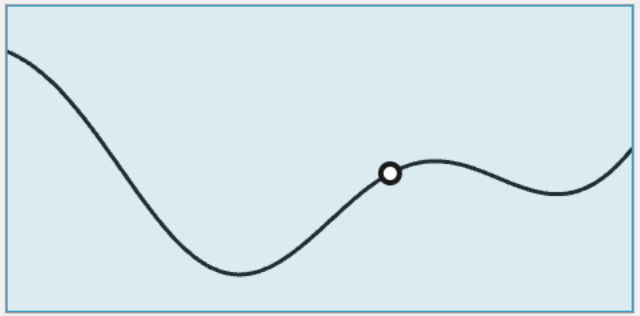
$$\mathcal{L}(w) \approx \mathcal{L}(v) + \nabla\mathcal{L}(v)(w - v) + \frac{1}{2}(w - v)^\top \boldsymbol{H}(v)(w - v)$$

in one dimension: $\mathcal{L}(w) \approx \mathcal{L}(v) + \mathcal{L}'(v)(w - v) + \frac{1}{2}\mathcal{L}''(v)(w - v)^2$

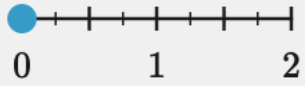
zoomed in on v: drag to move v



the whole function



$log_{10}\ zoom = 0.00$



zoom ×1.0: largest gap in view first order 4.84

Assumption 2: Lipschitz Smoothness

Definition 5.2.1. Lipschitz Continuous Gradient A function $\mathcal{L}(\theta) : \mathbb{R}^n \to \mathbb{R}$ has a Lipschitz continuous gradient if there exists a constant $L > 0$ such that for all $\boldsymbol{w}, \boldsymbol{v} \in \mathbb{R}^n$, the following inequality holds:

$$\|\nabla \mathcal{L}(\boldsymbol{w}) - \nabla \mathcal{L}(\boldsymbol{v})\| \leq L\|\boldsymbol{w} - \boldsymbol{v}\|.$$

The constant L is called the *Lipschitz constant*, and it bounds the rate of change of the gradient.

Using our assumptions

Here is what we have assumed about the function (a very mild assumption):

$$\mathcal{L}(\boldsymbol{w}) \approx \mathcal{L}(\boldsymbol{v}) + \nabla\mathcal{L}(\boldsymbol{v})(\boldsymbol{w}-\boldsymbol{v}) + \frac{1}{2}(\boldsymbol{w}-\boldsymbol{v})^\top \boldsymbol{H}(\boldsymbol{v})(\boldsymbol{w}-\boldsymbol{v})$$

$$\|\nabla\mathcal{L}(\boldsymbol{w}) - \nabla\mathcal{L}(\boldsymbol{v})\| \leq L\,\|\boldsymbol{w}-\boldsymbol{v}\|$$

How can these two things work together?

Using our assumptions

$$\mathcal{L}(\boldsymbol{w}) \approx \mathcal{L}(\boldsymbol{v}) + \nabla\mathcal{L}(\boldsymbol{v})(\boldsymbol{w}-\boldsymbol{v}) + \frac{1}{2}(\boldsymbol{w}-\boldsymbol{v})^\top \boldsymbol{H}(\boldsymbol{v})(\boldsymbol{w}-\boldsymbol{v})$$

$$\|\nabla\mathcal{L}(\boldsymbol{w}) - \nabla\mathcal{L}(\boldsymbol{v})\| \leq L\,\|\boldsymbol{w}-\boldsymbol{v}\|$$

$$\mathcal{L}(\boldsymbol{w}) \leq \mathcal{L}(\boldsymbol{v}) + \nabla\mathcal{L}(\boldsymbol{v})(\boldsymbol{w}-\boldsymbol{v}) + \frac{L}{2}\|\boldsymbol{w}-\boldsymbol{v}\|^2$$

$\nabla\mathcal{L}$ is L-Lipschitz, so the curvature is at most L: $(\boldsymbol{w}-\boldsymbol{v})^\top \boldsymbol{H}(\boldsymbol{v})(\boldsymbol{w}-\boldsymbol{v}) \leq L\|\boldsymbol{w}-\boldsymbol{v}\|^2$. This is the *descent lemma*.

Substituting in our terms

$$\mathcal{L}(\boldsymbol{w}) \leq \mathcal{L}(\boldsymbol{v}) + \nabla\mathcal{L}(\boldsymbol{v})(\boldsymbol{w} - \boldsymbol{v}) + \frac{L}{2}\|\boldsymbol{w} - \boldsymbol{v}\|^2$$

$$\theta^{(t+1)} = \theta^{(t)} - \alpha\nabla\mathcal{L}(\theta^{(t)})^\top$$

$$\mathcal{L}(\theta^{(t+1)}) \leq \mathcal{L}(\theta^{(t)}) + \nabla\mathcal{L}(\theta^{(t)})(\theta^{(t+1)} - \theta^{(t)}) + \frac{L}{2}\left\|\theta^{(t+1)} - \theta^{(t)}\right\|^2$$

w: the new point $\theta^{(t+1)}$, ***v***: the current point $\theta^{(t)}$

the descent lemma for one step of gradient descent

Combining to prove what we want

$$\theta^{(t+1)} = \theta^{(t)} - \alpha \nabla \mathcal{L}(\theta^{(t)})^\top$$

$$\mathcal{L}(\theta^{(t+1)}) \leq \mathcal{L}(\theta^{(t)}) + \nabla \mathcal{L}(\theta^{(t)})(\theta^{(t+1)} - \theta^{(t)}) + \frac{L}{2} \left\| \theta^{(t+1)} - \theta^{(t)} \right\|^2$$

How can we use the first equation to think about one of the terms in the second equation?

Combining to prove what we want

$$\theta^{(t+1)} = \theta^{(t)} - \alpha \nabla \mathcal{L}(\theta^{(t)})^\top$$

$$\mathcal{L}(\theta^{(t+1)}) \leq \mathcal{L}(\theta^{(t)}) + \nabla \mathcal{L}(\theta^{(t)})(\theta^{(t+1)} - \theta^{(t)}) + \frac{L}{2} \|\theta^{(t+1)} - \theta^{(t)}\|^2$$

How can we use the first equation to think about one of the terms in the second equation?

$$\theta^{(t+1)} - \theta^{(t)} = -\alpha \nabla \mathcal{L}(\theta^{(t)})^\top$$

subtract $\theta^{(t)}$ from both sides

Combining to prove what we want

$$\theta^{(t+1)} - \theta^{(t)} = -\alpha \nabla \mathcal{L}(\theta^{(t)})^\top$$

$$\mathcal{L}(\theta^{(t+1)}) \leq \mathcal{L}(\theta^{(t)}) + \nabla \mathcal{L}(\theta^{(t)})(\theta^{(t+1)} - \theta^{(t)}) + \frac{L}{2}\|\theta^{(t+1)} - \theta^{(t)}\|^2$$

$$\mathcal{L}(\theta^{(t+1)}) \leq \mathcal{L}(\theta^{(t)}) + \nabla \mathcal{L}(\theta^{(t)})(-\alpha \nabla \mathcal{L}(\theta^{(t)})^\top) + \frac{L}{2}\|-\alpha \nabla \mathcal{L}(\theta^{(t)})^\top\|^2$$

both differences $\theta^{(t+1)} - \theta^{(t)}$ replaced by the step $-\alpha \nabla \mathcal{L}(\theta^{(t)})^\top$, in the first-order term and inside the norm

Combining to prove what we want

$$\text{Let } \alpha = \frac{1}{L},$$

$$\mathcal{L}(\theta^{(t+1)}) \leq \mathcal{L}(\theta^{(t)}) + \nabla\mathcal{L}(\theta^{(t)})\left(-\alpha\nabla\mathcal{L}(\theta^{(t)})^\top\right) + \frac{L}{2}\left\|-\alpha\nabla\mathcal{L}(\theta^{(t)})^\top\right\|^2$$

$$\mathcal{L}(\theta^{(t+1)}) \leq \mathcal{L}(\theta^{(t)}) - \frac{1}{L}\left\|\nabla\mathcal{L}(\theta^{(t)})\right\|^2 + \frac{L}{2}\left\|-\alpha\nabla\mathcal{L}(\theta^{(t)})^\top\right\|^2$$

$\nabla\mathcal{L}(\theta^{(t)})\ \nabla\mathcal{L}(\theta^{(t)})^\top = \|\nabla\mathcal{L}(\theta^{(t)})\|^2$ (a row vector times its own transpose), and $\alpha = 1/L$

Combining to prove what we want

$$\text{Let } \alpha = \frac{1}{L}, \quad \left\| -\alpha \nabla \mathcal{L}(\theta^{(t)})^\top \right\| = \frac{1}{L} \left\| \nabla \mathcal{L}(\theta^{(t)}) \right\|$$

$$\mathcal{L}(\theta^{(t+1)}) \leq \mathcal{L}(\theta^{(t)}) - \frac{1}{L} \left\| \nabla \mathcal{L}(\theta^{(t)}) \right\|^2 + \frac{L}{2} \left\| -\alpha \nabla \mathcal{L}(\theta^{(t)})^\top \right\|^2$$

$$\mathcal{L}(\theta^{(t+1)}) \leq \mathcal{L}(\theta^{(t)}) - \frac{1}{L} \left\| \nabla \mathcal{L}(\theta^{(t)}) \right\|^2 + \frac{L}{2} \cdot \frac{1}{L^2} \left\| \nabla \mathcal{L}(\theta^{(t)}) \right\|^2$$

$$= \mathcal{L}(\theta^{(t)}) - \frac{1}{2L} \left\| \nabla \mathcal{L}(\theta^{(t)}) \right\|^2$$

the length of the step: $\|-\alpha \nabla \mathcal{L}(\theta^{(t)})^\top\| = \alpha \|\nabla \mathcal{L}(\theta^{(t)})\| = (1/L)\|\nabla \mathcal{L}(\theta^{(t)})\|$, with $\alpha = 1/L$

the second-order term: $\|-\alpha \nabla \mathcal{L}(\theta^{(t)})^\top\|^2 = (1/L^2)\|\nabla \mathcal{L}(\theta^{(t)})\|^2$

combine: $-1/L + (L/2)(1/L^2) = -1/(2L)$

Progress bound!

We have now shown what we wanted to! If you assume your loss is twice differentiable and is smooth, then we know that each step improves proportional to the smoothness of the loss and the squared norm of the gradient.

$$\mathcal{L}(\theta^{(t+1)}) \leq \mathcal{L}(\theta^{(t)}) - \frac{1}{2L}\left\|\nabla \mathcal{L}(\theta^{(t)})\right\|^2$$

slide 7's goal, $\mathcal{L}(\theta^{(t+1)}) \leq \mathcal{L}(\theta^{(t)}) - c$, with $c = \frac{1}{2L}\left\|\nabla \mathcal{L}(\theta^{(t)})\right\|^2 \geq 0$

How would I ask about this?

One of the most frequent questions I get is: how will this appear on your exam?

How would I ask about this?

In class we saw the following, what is the largest value you can select for α such that we still make progress?

$$\text{Let } \alpha = \frac{1}{L}, \quad \left\| -\alpha \nabla \mathcal{L}(\theta^{(t)})^\top \right\| = \frac{1}{L} \left\| \nabla \mathcal{L}(\theta^{(t)}) \right\|$$

$$\mathcal{L}(\theta^{(t+1)}) \leq \mathcal{L}(\theta^{(t)}) - \frac{1}{L} \left\| \nabla \mathcal{L}(\theta^{(t)}) \right\|^2 + \frac{L}{2} \left\| -\alpha \nabla \mathcal{L}(\theta^{(t)})^\top \right\|^2$$

$$\mathcal{L}(\theta^{(t+1)}) \leq \mathcal{L}(\theta^{(t)}) - \frac{1}{2L} \left\| \nabla \mathcal{L}(\theta^{(t)}) \right\|^2$$

More interesting questions

We get closer every step, but:

- Do we ever reach the minimum?
- Under what conditions do we reach the minimum?
- How quickly do we reach the minimum?

To answer more specific questions like this, often our assumptions must also get more specific (or our tools much more powerful).

More interesting questions → more powerful assumptions

$$\min_{\theta \in T} \mathcal{L}(\theta)$$

The assumption we will make through the rest of this lecture is that the optimization we wish to solve is convex.

An optimization problem is convex if and only if:

- The function is convex
- The solution set (domain) is convex

Convex Optimization

$$\min_{\theta \in T} \mathcal{L}(\theta)$$

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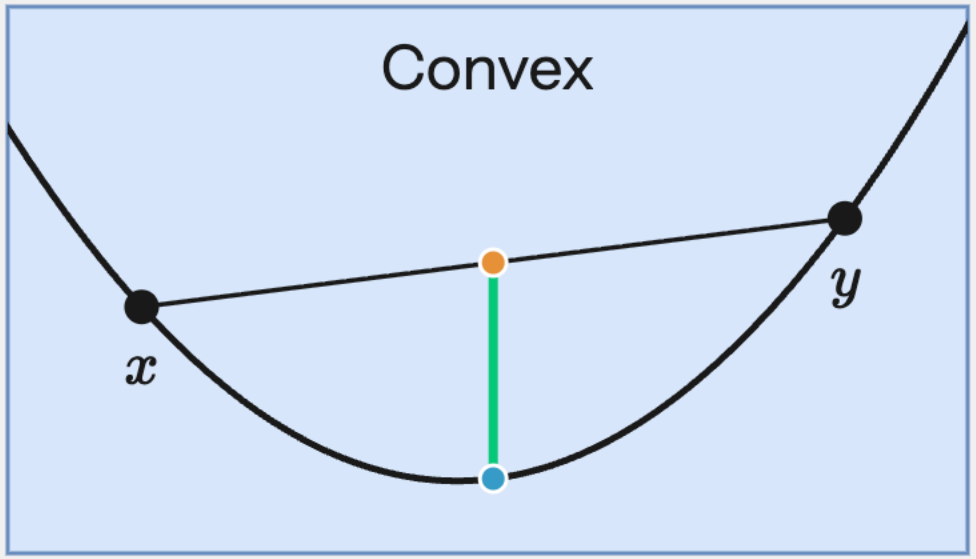
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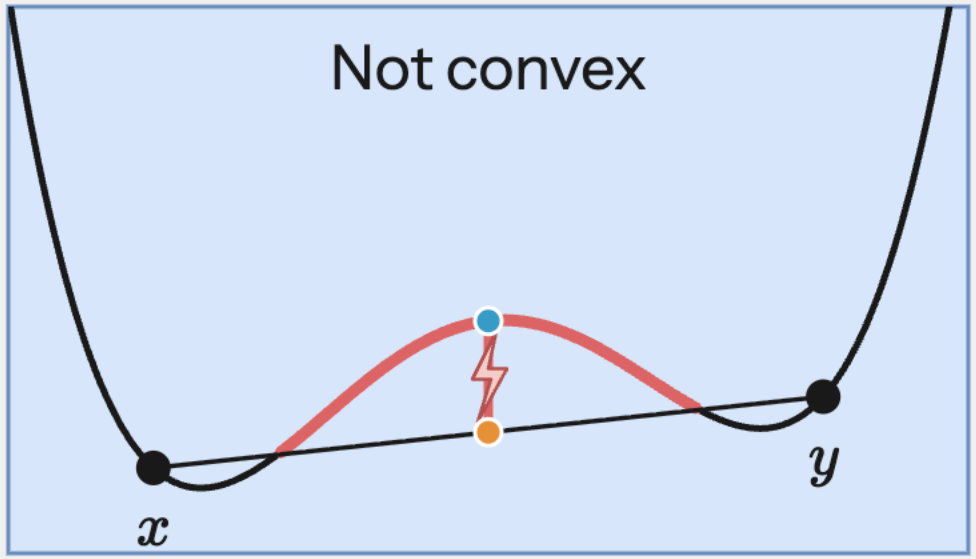
Convex Function

$$\forall x, y \in \mathrm{dom}(f), \quad \forall t \in [0, 1] :$$

$$f\big(tx + (1-t)y\big) \leq tf(x) + (1-t)f(y)$$



$f(tx + (1-t)y) = 0.4 < 1.77$



$f(tx + (1-t)y) = 1.4 > 0.7$

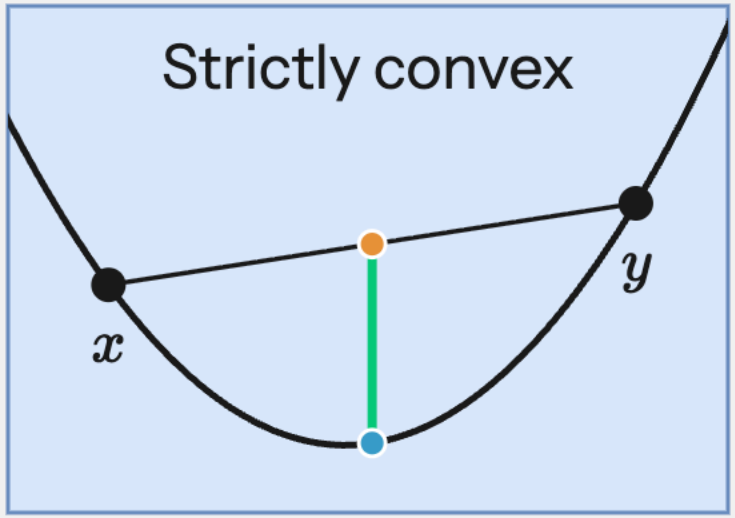
$t = 0.50$



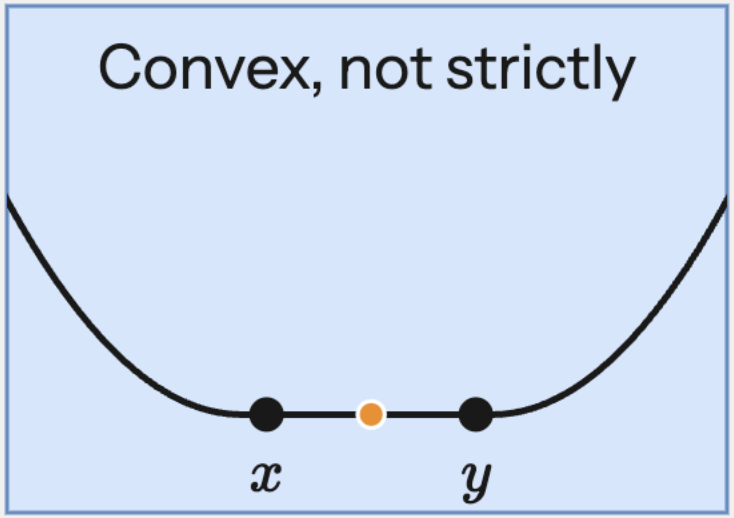
Strictly Convex Function

$$\forall x, y \in \mathrm{dom}(f),\ x \neq y,\ \ \forall t \in (0,1):$$

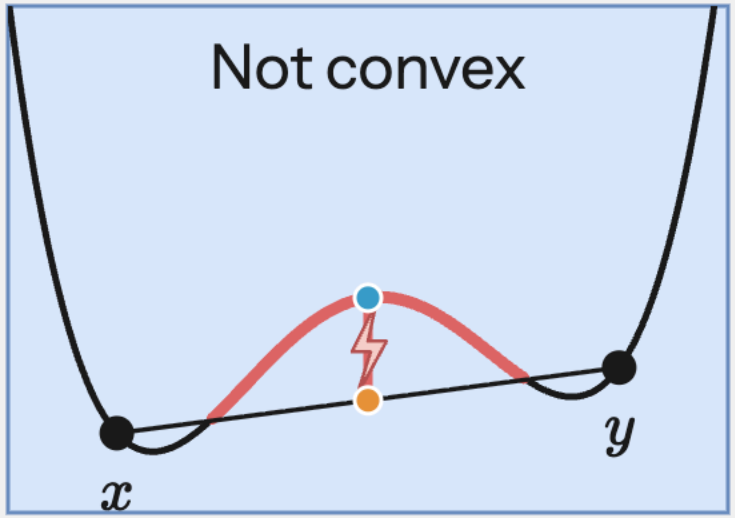
$$f\big(tx + (1-t)y\big) < tf(x) + (1-t)f(y)$$



$f(tx + (1-t)y) = 0.4 < 1.77$



$f(tx + (1-t)y) = 0.6 = 0.6$



$f(tx + (1-t)y) = 1.4 > 0.7$

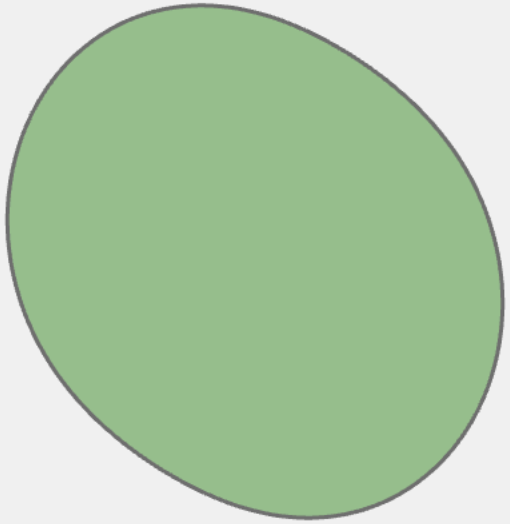
$t = 0.50$



Convex set

$$C \subseteq \mathbb{R}^n \text{ is convex if } \forall \boldsymbol{x}, \boldsymbol{y} \in C \text{ and } \forall t \in [0,1]:$$

$$t\boldsymbol{x} + (1-t)\boldsymbol{y} \in C$$

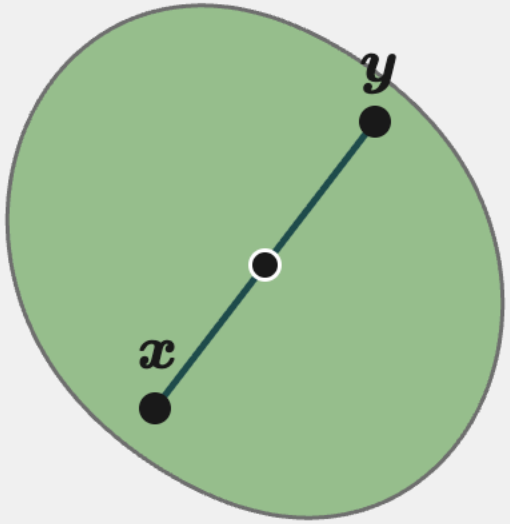




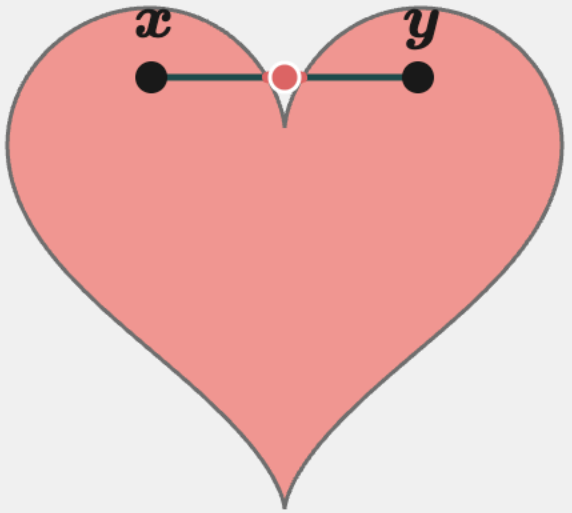
Convex set

$$C \subseteq \mathbb{R}^n \text{ is convex if } \forall \boldsymbol{x}, \boldsymbol{y} \in C \text{ and } \forall t \in [0,1]:$$

$$t\boldsymbol{x} + (1-t)\boldsymbol{y} \in C$$



the whole segment is in the set



part of the segment leaves the set

$t = 0.50$



Implications of Convexity

There are two critical implications of convexity (that are proven in the notes):

1. All local minimums are global minimums.
2. If a function is strictly convex, then the global minimum is unique.

Implications of Convexity

$$\theta^{(t+1)} = \theta^{(t)} - \alpha \nabla_\theta \mathcal{L}(\theta^{(t)})^\top$$

If we now assume that the optimization problem we seek to solve with gradient descent is convex (not necessarily strictly convex). We want to answer the questions:

- Do we ever ***reach*** the minimum?
- Under what conditions do we reach the minimum?

Convergence in Convexity

Whether or not we “reach” the optimum is thought about through the definition of convergence:

Definition 5.4.1. Convergence A sequence $x_1, x_2, \ldots$ is said to *converge* to a limit T if for any $\epsilon > 0$ we have an integer K such that $\forall M > K$, $|x_M - T| < \epsilon$.

To understand if this is the case, let’s turn to linear regression where we have a closed form solution for the limit *T* to which we would like to converge.

GD but in our linear model

Recalling our linear regression set up with GD:

$$f(\boldsymbol{x}, \theta) = \boldsymbol{x}^\top \theta, \qquad \hat{y} = f(\boldsymbol{x}, \theta)$$

$$\begin{aligned}
\theta^{(t+1)} &= \theta^{(t)} - \alpha \nabla_\theta \mathcal{L}\left(\boldsymbol{y}, f^{\theta^{(t)}}(\boldsymbol{X})\right)^\top \\
&= \theta^{(t)} - \alpha \nabla_\theta \left(\frac{1}{2} \left\| \boldsymbol{y} - \boldsymbol{X}\theta^{(t)} \right\|_2^2\right)^\top \\
&= \theta^{(t)} - \alpha \boldsymbol{X}^\top (\boldsymbol{X}\theta^{(t)} - \boldsymbol{y})
\end{aligned}$$

the loss of linear regression, the squared error: $\mathcal{L}(\theta) = \frac{1}{2}\|\boldsymbol{y} - \boldsymbol{X}\theta\|_2^2$

its gradient is $(\boldsymbol{X}\theta - \boldsymbol{y})^\top \boldsymbol{X}$ (lecture 02), so the transpose is $\boldsymbol{X}^\top(\boldsymbol{X}\theta - \boldsymbol{y})$

Benefits of tractability

Recalling our linear regression set up with GD:

$$\theta^{(t+1)} = \theta^{(t)} - \alpha \boldsymbol{X}^\top (\boldsymbol{X}\theta^{(t)} - \boldsymbol{y})$$

Moreover, recall that we would like to converge to:

$$\theta^* = (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{X}^\top \boldsymbol{y}$$

Using tools from calculus

A tool often used in calculus to think about convergence is recurrence relations where we would like to write out the terms of our sequence in relation to preceding terms:

$$\theta^{(t+1)} = \theta^{(t)} - \alpha \boldsymbol{X}^\top (\boldsymbol{X}\theta^{(t)} - \boldsymbol{y})$$

$$= (\boldsymbol{I} - \alpha \boldsymbol{X}^\top \boldsymbol{X})\theta^{(t)} + \alpha \boldsymbol{X}^\top \boldsymbol{y}$$

collect the terms in $\theta^{(t)}$: $\theta^{(t)} = \mathbf{I}\theta^{(t)}$

Let's understand what we are looking at

$$\theta^{(t+1)} = \theta^{(t)} - \alpha \boldsymbol{X}^\top (\boldsymbol{X}\theta^{(t)} - \boldsymbol{y})$$

$$= \underbrace{(\boldsymbol{I} - \alpha \boldsymbol{X}^\top \boldsymbol{X})}\theta^{(t)} + \underbrace{\alpha \boldsymbol{X}^\top \boldsymbol{y}}$$

What are the types of these two objects?

Let's understand what we are looking at

$$\theta^{(t+1)} = \theta^{(t)} - \alpha \boldsymbol{X}^\top (\boldsymbol{X}\theta^{(t)} - \boldsymbol{y})$$

$$= \underbrace{(\boldsymbol{I} - \alpha \boldsymbol{X}^\top \boldsymbol{X})}\theta^{(t)} + \underbrace{\alpha \boldsymbol{X}^\top \boldsymbol{y}}$$

What are the types of these two objects?

$(\boldsymbol{I} - \alpha \boldsymbol{X}^\top \boldsymbol{X}) \in \mathbb{R}^{n \times n}$ $\alpha \boldsymbol{X}^\top \boldsymbol{y} \in \mathbb{R}^{n \times 1}$

Now we can think about the form

Given the types of the operations in our previous slide we can see that this is an affine recurrence relation rather than a linear one. As is, our recurrence is of the form:

$$\theta^{(t+1)} = \boldsymbol{B}\theta^{(t)} + \boldsymbol{c}, \qquad t \geq 0$$

Now we can think about the form

Given the types of the operations in our previous slide we can see that this is an affine recurrence relation rather than a linear one. As is, our recurrence is of the form:

$$\theta^{(t+1)} = \boldsymbol{B}\theta^{(t)} + \boldsymbol{c}, \qquad t \geq 0$$

With this relation to get from our initialization to our end point we have a recurrence of the form:

$$\theta^{(3)} = \boldsymbol{B}\Big(\boldsymbol{B}\big(\boldsymbol{B}\theta^{(0)} + \boldsymbol{c}\big) + \boldsymbol{c}\Big) + \boldsymbol{c}$$ three steps

Improving the form

A nicer form of the recurrence would be to get it in the form:

$$\theta^{(t+1)} = \boldsymbol{A}(\theta^{(t)} + \boldsymbol{\beta}) - \boldsymbol{\beta}, \qquad \text{for some } \boldsymbol{A}, \boldsymbol{\beta}.$$

In particular because if we can convert it to the above affine transformation we have that the recurrence is:

$$\theta^{(3)} = \boldsymbol{A}^3(\theta^{(0)} + \boldsymbol{\beta}) - \boldsymbol{\beta}$$

what is left: ***A*** three times, then one **−β**

$$\theta^{(t)} = \boldsymbol{A}^t(\theta^{(0)} + \boldsymbol{\beta}) - \boldsymbol{\beta}$$

after *t* steps

A bit of algebra to get where we want…

$$\boldsymbol{A}(\theta^{(t)} + \boldsymbol{\beta}) - \boldsymbol{\beta}$$

$$= \boldsymbol{A}\theta^{(t)} + (\boldsymbol{A} - \boldsymbol{I})\boldsymbol{\beta}$$

expand, and collect the **β** terms

$$\Longrightarrow \quad \boldsymbol{A} = \boldsymbol{B}, \qquad \boldsymbol{\beta} = (\boldsymbol{B} - \boldsymbol{I})^{-1}\boldsymbol{c}$$

match with $\theta^{(t+1)} = \boldsymbol{B}\theta^{(t)} + \boldsymbol{c}$: the terms in $\theta^{(t)}$ give $\boldsymbol{A} = \boldsymbol{B}$, the rest $(\boldsymbol{A} - \mathbf{I})\boldsymbol{\beta} = \boldsymbol{c}$

A bit of algebra to get where we want...

$$A(\theta^{(t)} + \beta) - \beta$$

$$= A\theta^{(t)} + (A - I)\beta$$

$$\implies A = B, \qquad \beta = (B - I)^{-1}c$$

$$= \underbrace{(I - \alpha X^\top X)}_{B}\theta^{(t)} + \underbrace{\alpha X^\top y}_{c}$$

our update (slide 35)

A bit of algebra to get where we want...

$$\boldsymbol{A} = \boldsymbol{B}, \qquad \boldsymbol{\beta} = (\boldsymbol{B} - \boldsymbol{I})^{-1}\boldsymbol{c}$$

$$= \underbrace{(\boldsymbol{I} - \alpha \boldsymbol{X}^\top \boldsymbol{X})}_{\boldsymbol{B}}\theta^{(t)} + \underbrace{\alpha \boldsymbol{X}^\top \boldsymbol{y}}_{\boldsymbol{c}}$$

$$\boldsymbol{A} = (\boldsymbol{I} - \alpha \boldsymbol{X}^\top \boldsymbol{X}) \qquad \boldsymbol{A} = \boldsymbol{B}$$

$$\boldsymbol{\beta} = \left((\boldsymbol{I} - \alpha \boldsymbol{X}^\top \boldsymbol{X}) - \boldsymbol{I}\right)^{-1} \alpha \boldsymbol{X}^\top \boldsymbol{y} \qquad \boldsymbol{\beta} = (\boldsymbol{B} - \boldsymbol{I})^{-1}\boldsymbol{c}$$

Something interesting happened!

$$\boldsymbol{A} = (\boldsymbol{I} - \alpha \boldsymbol{X}^\top \boldsymbol{X})$$

$$\boldsymbol{\beta} = -(\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{X}^\top \boldsymbol{y} = -\theta^*$$

$$\theta^{(t)} = \boldsymbol{A}^t (\theta^{(0)} + \boldsymbol{\beta}) - \boldsymbol{\beta}$$ slide 40

the least-squares solution of lecture 02 (slide 34): **β** = −θ*

Something interesting happened!

$$\boldsymbol{A} = (\boldsymbol{I} - \alpha \boldsymbol{X}^\top \boldsymbol{X})$$

$$\boldsymbol{\beta} = -(\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{X}^\top \boldsymbol{y} = -\theta^*$$

$$\theta^{(t)} = \boldsymbol{A}^t (\theta^{(0)} + \boldsymbol{\beta}) - \boldsymbol{\beta}$$

$$\theta^{(t)} = (\boldsymbol{I} - \alpha \boldsymbol{X}^\top \boldsymbol{X})^t (\theta^{(0)} - \theta^*) + \theta^*$$

substitute $\boldsymbol{A} = \boldsymbol{I} - \alpha \boldsymbol{X}^\top \boldsymbol{X}$ and $\boldsymbol{\beta} = -\theta^*$

When do we converge?

$$\theta^{(t)} = (\boldsymbol{I} - \alpha \boldsymbol{X}^\top \boldsymbol{X})^t (\theta^{(0)} - \theta^*) + \theta^*$$

Under what conditions does this give us what we would like as a result?

When do we converge?

$$\theta^{(t)} = (\boldsymbol{I} - \alpha \boldsymbol{X}^\top \boldsymbol{X})^t (\theta^{(0)} - \theta^*) + \theta^*$$

Under what conditions does this give us what we would like as a result?

$$(\boldsymbol{I} - \alpha \boldsymbol{X}^\top \boldsymbol{X})^t (\theta^{(0)} - \theta^*) \to \mathbf{0}$$

$$(\boldsymbol{I} - \alpha \boldsymbol{X}^\top \boldsymbol{X})^t \to \mathbf{0}$$

$\theta^{(t)} \to \theta^*$ exactly when the first term vanishes

for every start $\theta^{(0)}$: the matrix power itself must vanish

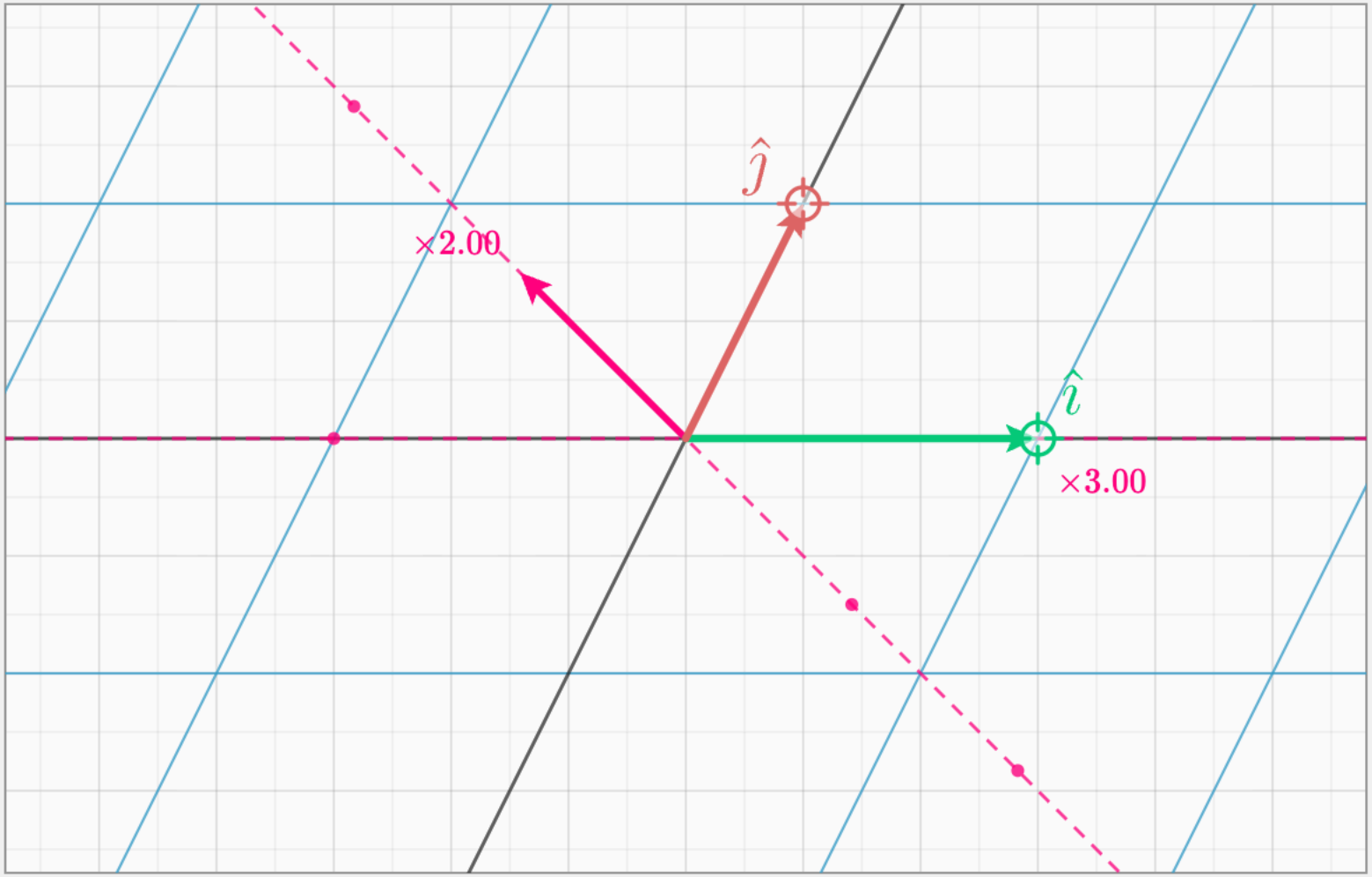
Eigenvalues and eigenvectors

Definition. Eigenvalues and eigenvectors Let $\boldsymbol{A} \in \mathbb{R}^{n \times n}$ be a square matrix. A scalar λ is an *eigenvalue* of $\boldsymbol{A}$ if there is a non-zero vector $\boldsymbol{u} \in \mathbb{R}^n$ such that

$$\boldsymbol{A}\boldsymbol{u} = \lambda \boldsymbol{u}.$$

Such a vector $\boldsymbol{u}$ is an *eigenvector* of $\boldsymbol{A}$ for the eigenvalue λ: $\boldsymbol{A}$ only scales it, by λ.

Visualising eigenvectors



$$A = \begin{bmatrix} 3.00 & 1.00 \\ 0.00 & 2.00 \end{bmatrix}$$

$\lambda_1 = 3.00,\ \boldsymbol{u}_1 = (1.00, 0.00)$
$\lambda_2 = 2.00,\ \boldsymbol{u}_2 = (-0.71, 0.71)$

$t = 1.00$

0 0.5 1

When do we converge?

$$(\boldsymbol{I} - \alpha \boldsymbol{X}^\top \boldsymbol{X})^t \to \mathbf{0}$$

If all the eigenvalues are between −1 and 1, this will converge to **0**. Formally:

Convergence in Convexity

$$(\boldsymbol{I} - \alpha \boldsymbol{X}^\top \boldsymbol{X})^t \to \mathbf{0}$$

If all the eigenvalues are between −1 and 1, this will converge to **0**. Formally:

$$\lambda_{\min}(\boldsymbol{A})\|\boldsymbol{x}\|_2^2 \leq \boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x} \leq \lambda_{\max}(\boldsymbol{A})\|\boldsymbol{x}\|_2^2$$

for a symmetric matrix A; equality along its eigenvectors

Convergence in Convexity

$$\|\theta^{(t)} - \theta^*\|_2^2 = \left\|(\boldsymbol{I} - \alpha \boldsymbol{X}^\top \boldsymbol{X})^t (\theta^{(0)} - \theta^*)\right\|_2^2$$

the squared distance to θ*, from slide 46

$$= (\theta^{(0)} - \theta^*)^\top (\boldsymbol{I} - \alpha \boldsymbol{X}^\top \boldsymbol{X})^{2t} (\theta^{(0)} - \theta^*)$$

$\|\boldsymbol{M}\boldsymbol{v}\|^2 = \boldsymbol{v}^\top \boldsymbol{M}^\top \boldsymbol{M}\boldsymbol{v} = \boldsymbol{v}^\top \boldsymbol{M}^2 \boldsymbol{v}$, as $\boldsymbol{M} = \mathbf{I} - \alpha \boldsymbol{X}^\top \boldsymbol{X}$ is symmetric

$$\|\theta^{(t)} - \theta^*\|_2^2 \leq \lambda_{\max}\left((\boldsymbol{I} - \alpha \boldsymbol{X}^\top \boldsymbol{X})^{2t}\right) \|\theta^{(0)} - \theta^*\|_2^2$$

$$\|\theta^{(t)} - \theta^*\|_2^2 \geq \lambda_{\min}\left((\boldsymbol{I} - \alpha \boldsymbol{X}^\top \boldsymbol{X})^{2t}\right) \|\theta^{(0)} - \theta^*\|_2^2$$

the bound of slide 51 with $A = (\mathrm{I} - \alpha \mathrm{X}^\top \mathrm{X})^{2t}$ and $x = \theta^{(0)} - \theta^*$

Convergence in Convexity

$$\|\theta^{(t)} - \theta^*\|_2^2 \leq \lambda_{\max}\left((\boldsymbol{I} - \alpha \boldsymbol{X}^\top \boldsymbol{X})^{2t}\right)\|\theta^{(0)} - \theta^*\|_2^2$$

$$\|\theta^{(t)} - \theta^*\|_2^2 \geq \lambda_{\min}\left((\boldsymbol{I} - \alpha \boldsymbol{X}^\top \boldsymbol{X})^{2t}\right)\|\theta^{(0)} - \theta^*\|_2^2$$

$$\lambda_{\max} < 1: \quad \text{always converge}$$

$$\lambda_{\min} \geq 1: \quad \text{always diverge}$$

$$\lambda_{\min} < 1 \text{ but } \lambda_{\max} \geq 1: \quad \text{depends on } \theta^{(0)}$$

λ: the eigenvalues of $(\boldsymbol{I} - \alpha \boldsymbol{X}^\top \boldsymbol{X})^2$; those of the $2t$-th power are their t-th powers

Convergence in Convexity

$$\|\theta^{(t)} - \theta^*\|_2^2 \leq \lambda_{\max}\left((\boldsymbol{I} - \alpha \boldsymbol{X}^\top \boldsymbol{X})^{2t}\right)\|\theta^{(0)} - \theta^*\|_2^2$$

$$\|\theta^{(t)} - \theta^*\|_2^2 \geq \lambda_{\min}\left((\boldsymbol{I} - \alpha \boldsymbol{X}^\top \boldsymbol{X})^{2t}\right)\|\theta^{(0)} - \theta^*\|_2^2$$

How can we influence the eigenvalues?

$$\lambda_{\max} < 1: \quad \text{always converge}$$

$$\lambda_{\min} \geq 1: \quad \text{always diverge}$$

$$\lambda_{\min} < 1 \text{ but } \lambda_{\max} \geq 1: \quad \text{depends on } \theta^{(0)}$$

Choosing the learning rate

$$\boldsymbol{X}^\top \boldsymbol{X}\, \boldsymbol{u} = \lambda \boldsymbol{u}$$

$$(\boldsymbol{I} - \alpha \boldsymbol{X}^\top \boldsymbol{X})\boldsymbol{u} = (1 - \alpha\lambda)\, \boldsymbol{u}$$

$$(\boldsymbol{I} - \alpha \boldsymbol{X}^\top \boldsymbol{X})^2\boldsymbol{u} = (1 - \alpha\lambda)^2\, \boldsymbol{u}$$

u: an eigenvector of $\boldsymbol{X}^\top\boldsymbol{X}$, $\lambda \geq 0$ its eigenvalue ($\boldsymbol{X}^\top\boldsymbol{X}$ is PSD)

$\boldsymbol{u} - \alpha\boldsymbol{X}^\top\boldsymbol{X}\boldsymbol{u} = \boldsymbol{u} - \alpha\lambda\boldsymbol{u}$: so $1 - \alpha\lambda$ is an eigenvalue of $\boldsymbol{I} - \alpha\boldsymbol{X}^\top\boldsymbol{X}$

applied twice: $(1 - \alpha\lambda)^2$ is an eigenvalue of $(\boldsymbol{I} - \alpha\boldsymbol{X}^\top\boldsymbol{X})^2$

$$(1 - \alpha\lambda)^2 < 1 \iff -1 < 1 - \alpha\lambda < 1 \iff 0 < \alpha\lambda < 2$$

"always converge" (slide 53) needs every $(1 - \alpha\lambda)^2 < 1$

$$\text{for every eigenvalue } \lambda \text{ of } \boldsymbol{X}^\top\boldsymbol{X}: \qquad \alpha < \frac{2}{\lambda_{\max}(\boldsymbol{X}^\top\boldsymbol{X})}$$

the largest eigenvalue is the binding one (assuming $\boldsymbol{X}^\top\boldsymbol{X}$ invertible, so every $\lambda > 0$)

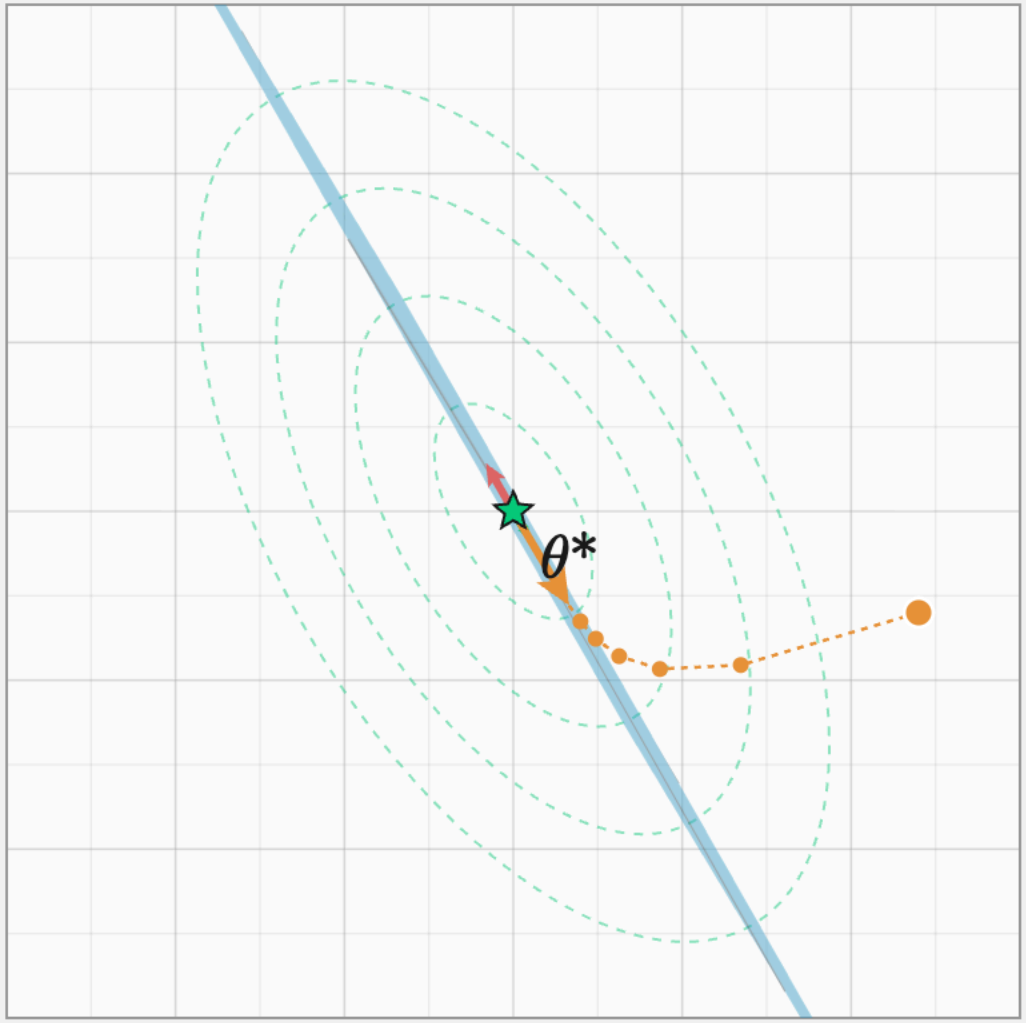
Convergence in Convexity

$$\alpha < \frac{2}{\lambda_{\max}(\boldsymbol{X}^\top \boldsymbol{X})}$$

$$\lambda_{\max} < 1: \quad \text{always converge}$$

For least squares the Lipschitz constant of $\nabla\mathcal{L}$ is $L = \lambda_{\max}(\boldsymbol{X}^\top \boldsymbol{X})$: this is $\alpha < 2/L$, the same as for the progress bound (slide 22)

Gradient descent as a linear transformation



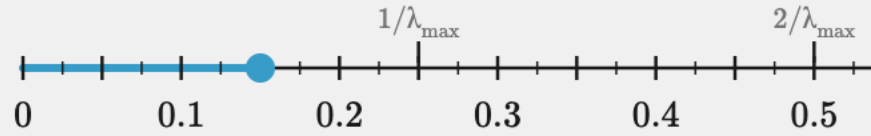
$$A = I - \alpha X^\top X = \begin{bmatrix} 0.51 & -0.19 \\ -0.19 & 0.74 \end{bmatrix}$$

$$\theta^{(t)} - \theta^* = A^t(\theta^{(0)} - \theta^*)$$

$X^\top X$: eigenvalues $\lambda_1 = 4,\ \lambda_2 = 1$ (eigenvectors pink)

A: eigenvalues $1 - 4\alpha = \mathbf{0.40},\ 1 - \alpha = \mathbf{0.85}$

$\alpha = 0.150$



$t = 6$: $\|\theta^{(t)} - \theta^*\| = 0.649$

Feedback



https://forms.gle/261FWs7hSWDARr9W7

Please let me know your thoughts and concerns!